

Before class: sharing electrons

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Read the cards first, then answer the questions without notes. A periodic table and the answer key are at the back.

1 · Read first (5 min)

From IC1 to IC2

In IC1 you saw atoms reach a stable **octet** (eight valence electrons) by **transferring** electrons. Metals lose electrons, nonmetals gain them, and the ions attract: an **ionic bond**.

When two **nonmetals** meet, neither gives up electrons easily. Instead they **share** them. A shared pair of electrons is a **covalent bond**, and a group of atoms joined by covalent bonds is a **molecule**. Water (H₂O), methane (CH₄) and cholesterol are all made of molecules.

Words you will meet in the lecture

- **Bonding pair**: two electrons shared between two atoms (drawn as a dash, H-H).
- **Lone pair**: two valence electrons on one atom that are not shared.
- **Double / triple bond**: two or three shared pairs between the same two atoms.
- **Lewis structure**: a drawing of all bonding pairs and lone pairs in a molecule.
- **Electronegativity**: how strongly an atom pulls on shared electrons.
- **Polar**: having a slightly positive end and a slightly negative end.
- **Intermolecular forces**: attractions *between* molecules.

A tool you already have: valence electrons

For main-group elements, the number of valence electrons equals the **group number**: H 1 · C (4A) 4 · N (5A) 5 · O (6A) 6 · F, Cl (7A) 7. Hydrogen is happy with just **two** electrons (the duet rule).

2 · Readiness check

Ten quick questions on what you need for today.

Q1 IC1 recap

How many valence electrons does an oxygen atom (Group 6A) have?

- A. 2
- B. 4
- C. 6
- D. 8

Q2 IC1 recap

A chlorine atom (Group 7A) has ____ valence electrons.

Q3 IC2-LO1

Which pair of elements is most likely to form a **covalent** bond?

- A. Na and Cl
- B. K and O
- C. N and H
- D. Ca and F

Q4 IC2-LO1

In a covalent bond, electrons are transferred completely from one atom to another.

True / False

Q5 IC2-LO1

One shared pair (a single covalent bond) contains how many electrons?

- A. 1
- B. 2
- C. 4
- D. 8

Q6 IC2-LO1

A hydrogen atom is stable with how many electrons in its valence shell?

- A. 1
- B. 2
- C. 8
- D. 18

Q7 IC2-LO2

Classify each compound.

Options: ionic (metal + nonmetal) · covalent (nonmetals only)

NaCl

H₂O

CO₂

MgO

Q8 IC2-LO1

Carbon (Group 4A) needs ____ more electrons to reach an octet, so it usually forms that many covalent bonds.

Q9 IC2-LO2

Ammonium chloride, NH_4Cl , contains:

- A. ionic bonds only
- B. covalent bonds only
- C. both ionic and covalent bonds
- D. no bonds

Q10 IC2-LO3

A lone pair is a pair of valence electrons that is not shared with another atom.

True / False

3 · Think ahead

No wrong answers here. Make a guess, then check it after the lecture (Parts 4–6).

Q11 IC2-LO6

Water (H_2O , 18 amu) boils at 100 °C. Methane (CH_4 , 16 amu) boils at about –161 °C. The molecules have almost the same mass. Why might water boil so much higher?

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Periodic table · each box: atomic number, symbol, atomic mass, **electronegativity** · colours: metal, metalloid, nonmetal, noble gas

1A 1	2A 2	3	4	5	6	7	8	9	10	11	12	3A 13	4A 14	5A 15	6A 16	7A 17	8A 18
1 H 1.01 2.1																	2 He 4.00 —
3 Li 6.94 1.0	4 Be 9.01 1.5											5 B 10.81 2.0	6 C 12.01 2.5	7 N 14.01 3.0	8 O 16.00 3.5	9 F 19.00 4.0	10 Ne 20.18 —
11 Na 22.99 0.9	12 Mg 24.31 1.2											13 Al 26.98 1.5	14 Si 28.09 1.8	15 P 30.97 2.1	16 S 32.07 2.5	17 Cl 35.45 3.0	18 Ar 39.95 —
19 K 39.10 0.8	20 Ca 40.08 1.0	21 Sc 44.96 1.3	22 Ti 47.87 1.5	23 V 50.94 1.6	24 Cr 52.00 1.6	25 Mn 54.94 1.5	26 Fe 55.85 1.8	27 Co 58.93 1.9	28 Ni 58.69 1.9	29 Cu 63.55 1.9	30 Zn 65.38 1.6	31 Ga 69.72 1.6	32 Ge 72.63 1.8	33 As 74.92 2.0	34 Se 78.97 2.4	35 Br 79.90 2.8	36 Kr 83.80 —
37 Rb 85.47 0.8	38 Sr 87.62 1.0	39 Y 88.91 1.2	40 Zr 91.22 1.4	41 Nb 92.91 1.6	42 Mo 95.95 1.8	43 Tc [98] 1.9	44 Ru 101.07 2.2	45 Rh 102.91 2.2	46 Pd 106.42 2.2	47 Ag 107.87 1.9	48 Cd 112.41 1.7	49 In 114.82 1.7	50 Sn 118.71 1.8	51 Sb 121.76 1.9	52 Te 127.60 2.1	53 I 126.90 2.5	54 Xe 131.29 —
55 Cs 132.91 0.7	56 Ba 137.33 0.9	57–71	72 Hf 178.49 1.3	73 Ta 180.95 1.5	74 W 183.84 1.7	75 Re 186.21 1.9	76 Os 190.23 2.2	77 Ir 192.22 2.2	78 Pt 195.08 2.2	79 Au 196.97 2.4	80 Hg 200.59 1.9	81 Tl 204.38 1.8	82 Pb 207.2 1.9	83 Bi 208.98 1.9	84 Po [209] 2.0	85 At [210] 2.2	86 Rn [222] —
87 Fr [223] 0.7	88 Ra [226] 0.9	89–103	104 Rf [267] —	105 Db [268] —	106 Sg [269] —	107 Bh [270] —	108 Hs [277] —	109 Mt [278] —	110 Ds [281] —	111 Rg [282] —	112 Cn [285] —	113 Nh [286] —	114 Fl [289] —	115 Mc [290] —	116 Lv [293] —	117 Ts [294] —	118 Og [294] —
Lanthanides		57 La 138.91 —	58 Ce 140.12 —	59 Pr 140.91 —	60 Nd 144.24 —	61 Pm [145] —	62 Sm 150.36 —	63 Eu 151.96 —	64 Gd 157.25 —	65 Tb 158.93 —	66 Dy 162.50 —	67 Ho 164.93 —	68 Er 167.26 —	69 Tm 168.93 —	70 Yb 173.05 —	71 Lu 174.97 —	
Actinides		89 Ac [227] 1.1	90 Th 232.04 1.3	91 Pa 231.04 1.4	92 U 238.03 1.4	93 Np [237] —	94 Pu [244] —	95 Am [243] —	96 Cm [247] —	97 Bk [247] —	98 Cf [251] —	99 Es [252] —	100 Fm [257] —	101 Md [258] —	102 No [259] —	103 Lr [262] —	

Atomic masses to 2 d.p. ([n] = mass number of the most stable isotope). Electronegativity: Pauling values as in Ball, *The Basics of GOB Chemistry*, Figure 4.4.3 (La–Lu 1.0–1.2; noble gases have no value).

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About 15 minutes. Read the three short cards, then try the readiness check. Wrong answers are fine: they show what to listen for in the lecture.

Answer key

2 · Readiness check

Q1 IC1 recap

How many valence electrons does an oxygen atom (Group 6A) have?

- A. 2
- B. 4
- C. 6
- D. 8

Group 6A → 6 valence electrons.

Q2 IC1 recap

A chlorine atom (Group 7A) has ____ valence electrons.

7

Group 7A → 7.

Q3 IC2-LO1

Which pair of elements is most likely to form a **covalent** bond?

- A. Na and Cl
- B. K and O
- C. N and H
- D. Ca and F

N and H are both nonmetals, so they share electrons. The others are metal + nonmetal (ionic).

Q4 IC2-LO1

In a covalent bond, electrons are transferred completely from one atom to another.

True / False → **False**

Transfer makes ions (ionic bond). A covalent bond **shares** electrons.

Q5 IC2-LO1

One shared pair (a single covalent bond) contains how many electrons?

A. 1

B. 2

C. 4

D. 8

A pair = 2 electrons.

Q6 IC2-LO1

A hydrogen atom is stable with how many electrons in its valence shell?

A. 1

B. 2

C. 8

D. 18

Its only shell holds two electrons: the duet rule.

Q7 IC2-LO2

Classify each compound.

NaCl → ionic (metal + nonmetal); H₂O → covalent (nonmetals only); CO₂ → covalent (nonmetals only); MgO → ionic (metal + nonmetal)

Q8 IC2-LO1

Carbon (Group 4A) needs ____ more electrons to reach an octet, so it usually forms that many covalent bonds.

4

4 + 4 = 8. That is why carbon forms four bonds, as in CH₄.

Q9 IC2-LO2

Ammonium chloride, NH_4Cl , contains:

- A. ionic bonds only
- B. covalent bonds only
- C. both ionic and covalent bonds**
- D. no bonds

N-H bonds inside NH_4^+ are covalent; NH_4^+ and Cl^- attract ionically.

Q10 IC2-LO3

A lone pair is a pair of valence electrons that is not shared with another atom.

True / False → **True**

Correct definition. Oxygen in water has two lone pairs.

3 • Think ahead

Q11 IC2-LO6

Water (H_2O , 18 amu) boils at 100 °C. Methane (CH_4 , 16 amu) boils at about –161 °C. The molecules have almost the same mass. Why might water boil so much higher?

Water molecules are **polar** (bent shape, O pulls electrons from H) and attract each other strongly through **hydrogen bonds**. Methane is **nonpolar**, so its molecules attract each other only through weak dispersion forces. More energy is needed to pull water molecules apart, so its boiling point is higher. You will see this in Parts 4 to 6.

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